Real Time Project Report

on

"CLEFFINITY"

Submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

in

ELECTRONICS AND COMMUNICATION ENGINEERING

by

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2024-25

DECLARATION

We hereby declare that the work described in this report, entitled "**CLEFFINITY**" which is being submitted by us in partial fulfillment for the award of the degree of **Bachelor of Technology** in the department of Electronics and Communication Engineering at **BVRIT HYDERABAD College of Engineering for Women (UGC Autonomous),** affiliated to **Jawaharlal Nehru Technological University Hyderabad**, Kukatpally, Hyderabad – 500085 is the result of original work carried out by us under the guidance of **Associate professor Mr. R Priyakanth.**

This work has not been submitted for any Degree/Diploma of this or any other institute/university to the best of our knowledge and belief.

Place: Hyderabad

Date:

Names and signatures of the students

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Certificate

This is to certify that the major/mini project report, entitled "**CLEFFINITY**" is a record of bonafide work carried out by **V.PRIYADARSHINI – 23WH1A0435, C. ASHWINI – 23WH1A0437, G. SARANYA – 23WH1A0439, G. TEJASWI – 23WH1A0458** in partial fulfillment for the award of the degree of **Bachelor of Technology** in the department of **Electronics and Communication Engineering** at **BVRIT HYDERABAD College of Engineering for Women (UGC Autonomous),** affiliated to **Jawaharlal Nehru Technological University Hyderabad,** Kukatpally, Hyderabad – 500085.

Supervisor **Head of the Department**

ACKNOWLEDGMENT

The satisfaction that accompanies in successful completion of the task would be incomplete without the mention of the people who made it possible.

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ABSTRACT

Music education has long relied on visual elements such as written notation and illustrations, posing significant challenges for individuals who are blind or visually impaired (VI). Traditional learning materials often fail to address the needs of these learners, creating barriers to musical literacy and self-expression.

Cleffinity is an innovative and inclusive tool designed to transform how blind and VI individuals engage with music. By shifting the focus from sight to sound and touch, Cleffinity enables learners to experience music in a more accessible and intuitive way. Its multisensory approach fosters independent exploration, allowing users to connect with musical concepts through tactile feedback and auditory cues.

More than just a learning aid, Cleffinity empowers users to develop confidence and creativity in their musical journey. It supports self-paced learning and encourages curiosity, making music education engaging and inclusive, whether in a classroom or at home. With Cleffinity, music becomes a space where everyone—regardless of visual ability—can participate, understand, and enjoy.

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1. Introduction

Music is a universal language that transcends boundaries and connects people across cultures. However, for individuals who are visually impaired, learning music can be an immense challenge. Traditional music education heavily depends on visual aids such as sheet music and diagrams, creating barriers for those who cannot see. Recognizing this issue, our project "Cleffinity" aims to provide an accessible and inclusive platform for music education.

Cleffinity is an assistive system that uses a combination of RFID technology, Braille, raised musical symbols, and audio feedback to help visually impaired individuals learn and experience music. With this tool, learners can feel musical notes on the page and hear them simultaneously, enabling a multisensory learning experience. Our goal is to empower visually impaired individuals with a self-guided, hands-free, and enjoyable method to explore music.

The system is designed to function with minimal effort on the part of the user. Once powered, Cleffinity detects the page being read by sensing the RFID tag embedded in it. The system instantly retrieves and plays the corresponding musical note or instruction from the microSD card through a speaker. This approach eliminates the need for any manual interface, making the device easy to operate by individuals of all age groups and skill levels. It can also serve as a foundational platform for more advanced music learning tools, enabling step-by-step progress for the user.

This project was inspired by the real-world challenges faced by blind learners who have limited access to musical notation and theory. While there are existing technologies that support text-to-speech or braille music transcription, they often require expensive equipment or specialized training. Cleffinity, in contrast, is affordable, easy to use, and scalable. It utilizes widely available components and open-source technology, which makes it suitable for deployment in schools, training centers, or even personal home use.

2. Objectives

- To develop an accessible system for music learning tailored to visually impaired users.
- To integrate tactile (Braille and raised symbols) and auditory (sound playback) feedback.
- To automate music note detection using RFID technology.
- To support multiple instrument tones for better auditory learning.
- To create a user-friendly, interactive, and inclusive learning tool.
- To explore the fusion of embedded systems and inclusive education for broader societal impact.

3. System Overview and Methodology

3.1 System Architecture

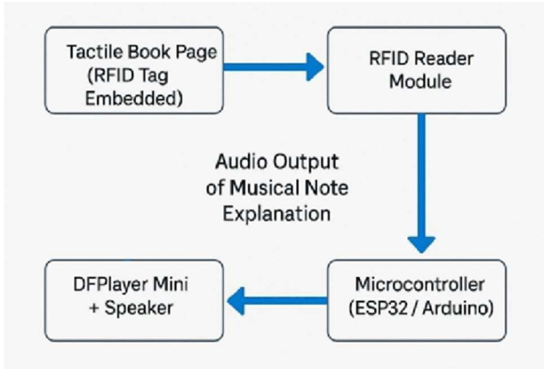
- RFID tag: Embedded on each page
- RFID reader: Detects tag when page is flipped
- Arduino UNO: Processes tag data and triggers audio
- DFPlayer Mini: Plays MP3 files from SD card
- Speaker: Outputs the note corresponding to the detected tag

3.2 Working

When a user flips a page in the tactile music book, the RFID reader detects the embedded tag. This information is sent to the Arduino, which retrieves the corresponding MP3 file from the SD card through the DFPlayer Mini module. The note is then played aloud, allowing the user to associate the tactile symbol with the auditory note.

The RFID reader (RC522) operates at 13.56 MHz and communicates with the Arduino UNO via the SPI protocol. Each RFID tag is programmed with a unique identifier, which is linked to an audio file. The DFPlayer Mini is a compact MP3 module with an inbuilt amplifier, making it ideal for portable applications. The use of raised tactile symbols printed or embossed onto the music sheet helps the user trace and identify note positions. The integration of both touch and sound significantly enhances learning efficiency.

Diagram-1 (Block Diagram)



3.3 Hardware Components

- Arduino UNO: Microcontroller board for system control
- RFID Reader (RC522): Reads the RFID tag from the music page
- RFID Tags (stickers): Embedded in each page to encode unique audio instructions
- DFPlayer Mini MP3 Module: Retrieves and plays MP3 files from a microSD card
- Speaker: Converts electrical signals into audio output
- Micro SD Card with MP3 audio files: Stores audio samples for notes and tones
- Braille-enabled and raised symbol music pages: Provide tactile feedback to users

Audio files are recorded in high quality to ensure clarity and distinctiveness of each note.

3.4 Connections

1. RFID RC522 to Arduino

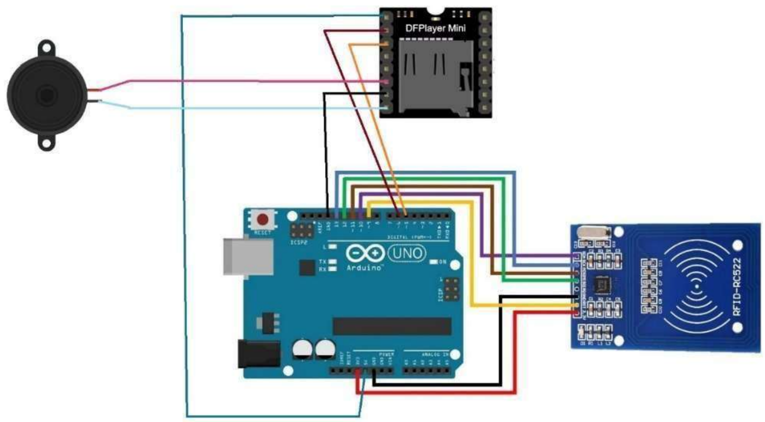
RFID RC522 Pin	Arduino Pin
SDA (SS)	10

SCK	13
MOSI	11
MISO	12
GND	GND
RST	9
3.3V	3.3V

2. DF Mini MP3 Player to Arduino

DF Mini MP3 Player Pin	Arduino Pin
VCC	5V
GND	GND
RX	D6
TX	D5
SPK1	Speaker (+)
SPK2	Speaker (−)

Diagram-2 (Circuit Diagram)



4. Features and Innovation

- **Touch and Sound Integration:** Users feel notes with their fingers and hear them simultaneously creating a powerful multisensory experience.
- **Multiple Instrument Sounds:** Each note is available in different instrument tones such as piano, violin, and guitar. This enhances auditory memory and improves the ability to differentiate sounds.
- **Hands-Free Operation:** The device activates automatically upon page flip, without any need for buttons or screen interaction.
- **Inclusive Learning:** Designed to be used by individuals with no prior exposure to music, ensuring accessibility for beginners and children.
- **Scalable and Customizable:** Can be adapted to various educational content beyond music, including alphabets, stories, and general knowledge.
- **Affordable and Open-Source:** Built using low-cost components and open-source tools to encourage widespread adoption.

5. Results and Testing

The Cleffinity prototype was assembled and tested in a controlled setting to verify the functionality of each component. The RFID module successfully detected tags placed on different pages, and the DFPlayer Mini module responded by playing the correct corresponding audio file for each tag.

All components—Arduino, RFID reader, speaker, and DFPlayer—worked in coordination to produce a smooth and consistent experience. The sound output was clear, and each tag triggered the expected audio file without errors. The system showed good responsiveness, with minimal delay between scanning a tag and hearing the audio.

The audio files were played through a speaker connected to a direct power source, ensuring consistent performance throughout testing. The logic for matching specific tags to particular audio files functioned as intended, confirming the accuracy of the software-hardware integration.

This successful test demonstrates that Cleffinity is technically functional and ready for further improvements or real-world application trials.

Diagram-3 (Connections)

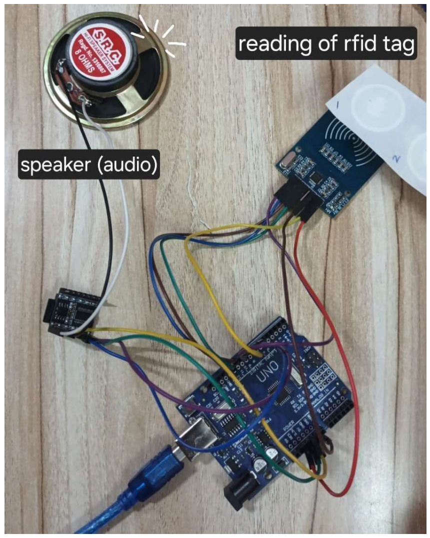
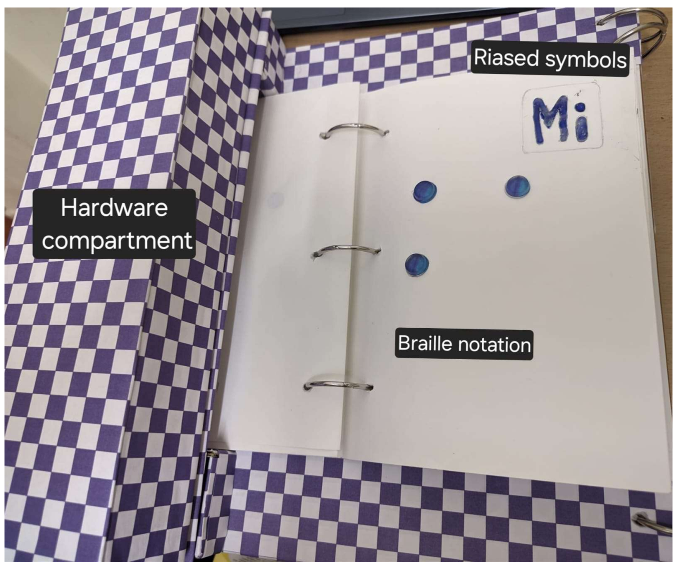


Diagram-4 (Total setup)



6. Conclusion

Cleffinity presents a step forward in inclusive education technology. By combining tactile and auditory feedback, it breaks down the barriers of traditional music learning for visually impaired individuals. The project showcases how embedded systems, when integrated thoughtfully, can transform education and accessibility.

The simplicity of Cleffinity allows for mass production and wide adoption. Its modularity means it can be adapted to include more complex musical pieces or even theoretical lessons in future versions. Additional features like Bluetooth connectivity, app integration, or quiz modes could further enhance the learning experience.

Future enhancements may include Bluetooth headphones, interactive quizzes, and support for music theory education. Overall, Cleffinity is not just a project—it's a movement toward making music truly for everyone.

7. References

Journals:

[1] J. A. Borges and D. Tomé, "Teaching music to blind children: New strategies for teaching through interactive use of musibraille software," Procedia Comput. Sci., vol. 27, pp. 19–27, 2014.

[2] L. Lu, K. A. Cochrane, J. Kang, and A. Girouard, "'Why are there so many steps?': Improving Access to Blind and Low Vision Music Learning through Personal Adaptations and Future Design Ideas," ACM Trans. Access. Comput., vol. 16, no. 2, pp. 1–20, 2023.

[3] N. F. Tisnawati, Yuliati, and E. Purbaningrum, "Braille innovation technology in teaching and learning process for visual impairment," JTP - J. Teknol. Pendidik., vol. 24, no. 2, pp. 224–235, 2022.

Weblinks:

1. **Arduino Documentation**
   https://www.arduino.cc/en/Guide/HomePage
   For understanding Arduino programming, board connections, and hardware interfacing.

2. **MFRC522 RFID Reader Library**
https://github.com/miguelbalboa/rfid
For using the MFRC522 RFID module with Arduino.

3. **DFPlayer Mini MP3 Player Library**
https://github.com/DFRobot/DFRobotDFPlayerMini
For interfacing DFPlayer Mini and playing audio files through Arduino.

4. **DFRobot Wiki - DFPlayer Mini**
https://wiki.dfrobot.com/DFPlayer_Mini_SKU_DFR0299
Technical documentation and example usage for the DFPlayer Mini module.

5. **Braille Music Notation Reference**
https://www.loc.gov/nls/music/
https://library.tsbvi.edu/assoc_files/92120013_1.pdf
Understanding how Braille is used for music education.

Appendix

Code:

```
#include <SPI.h>

#include <MFRC522.h>

#include <SoftwareSerial.h>p

#include <DFRobotDFPlayerMini.h>


#define SS_PIN 9

#define RST_PIN 8

#define DF_RX 6

#define DF_TX 7


MFRC522 rfid(SS_PIN, RST_PIN);

SoftwareSerial mySerial(DF_RX, DF_TX);

DFRobotDFPlayerMini myDFPlayer;


// Store 8 RFID tags and their assigned audio files

String tagList[8] = {

  "1d61baa4b1080", // Tag 1 -> 0001.mp3
```

```
  "1d62baa4b1080", // Tag 2 -> 0002.mp3   "1d63baa4b1080", //
Tag 3 -> 0003.mp3

  "1d64baa4b1080", // Tag 4 -> 0004.mp3

  "1d65baa4b1080", // Tag 5 -> 0005.mp3

  "1d66baa4b1080", // Tag 6 -> 0006.mp3

  "1d67baa4b1080", // Tag 7 -> 0007.mp3

"1d68baa4b1080"  // Tag 8 -> 0008.mp3

};



void setup() {

  Serial.begin(9600);

SPI.begin();

  rfid.PCD_Init();

  Serial.println("RFID Ready!");



  mySerial.begin(9600);

  Serial.println("Initializing DFPlayer...");



  if (!myDFPlayer.begin(mySerial)) {

    Serial.println("DFPlayer Mini NOT detected! Check wiring.");

    while (true);
```

```
 }


 Serial.println("DFPlayer Mini is working!");
 myDFPlayer.volume(15);  // Set volume (0-30)
}


void loop() {
 if (!rfid.PICC_IsNewCardPresent()) return;
 if (!rfid.PICC_ReadCardSerial()) return;


 // Read the card UID
String cardID = "";
 for (byte i = 0; i < rfid.uid.size; i++) {
   cardID += String(rfid.uid.uidByte[i], HEX);
 }
 Serial.print("Scanned Tag UID: ");
 Serial.println(cardID);


 // Check if the tag matches one of the 8 stored UIDs
 int assignedFile = -1;
 for (int i = 0; i < 8; i++) {
```

```
    if (cardID == tagList[i]) {
      assignedFile = i + 1; // Assigns file 0001.mp3 to 0008.mp3
      break;
    }
  }

  if (assignedFile != -1) {
    Serial.print("Playing Audio File ");
Serial.println(assignedFile);
    myDFPlayer.play(assignedFile);
  } else {
    Serial.println("Unknown Tag! No audio played.");
  }

  delay(4000);  // Wait before allowing another scan
  rfid.PICC_HaltA();
}
```

Final Prototype